

INTERNATIONAL STANDARD

**Fibre optic interconnecting devices and passive components – Performance standard –
Part 031-3: Non-connectorized single-mode 1×N and 2×N non-wavelength-selective branching devices for Category U – Uncontrolled environment**



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Fibre optic interconnecting devices and passive components – Performance standard –

Part 031-3: Non-connectorized single-mode 1×N and 2×N non-wavelength-selective branching devices for Category U – Uncontrolled environment

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INTERNATIONAL ELECTROTECHNICAL COMMISSION

**FIBRE OPTIC INTERCONNECTING
DEVICES AND PASSIVE COMPONENTS –
PERFORMANCE STANDARD –****Part 031-3: Non-connectorized single-mode 1×N and 2×N
non-wavelength-selective branching devices for Category U –
Uncontrolled environment**

FOREWORD

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International Standard IEC 61753-031-3 has been prepared by subcommittee 86B: Fibre optic interconnecting devices and passive components, of IEC technical committee 86: Fibre optics.

This second edition of IEC 61753-031-3 cancels and replaces the first edition published in 2009 and constitutes a technical revision.

This edition includes the following significant technical changes with respect to the previous edition:

- a) an updated and extended scope to reflect an introduction of two attenuation and uniformity performance classes for balanced NWBD instead of branching device technologies on the market;

- b) an updated list of normative references;
- c) a more simplified introduction to the two types of spectral bands;
- d) Clause 5, Performance requirements, has been updated and extended to reflect the introduction of two attenuation and uniformity performance classes for balanced NWBD;
- e) simplified test items to exclude tests for damp heat (steady state) and impact for performance requirements;
- f) Annexe A has been changed to introduce the calculated and minimum requirements for attenuation and uniformity;
- g) Annex B has been updated to reflect revised performance requirements.

The text of this standard is based on the following documents:

FDIS	Report on voting
86B/3792/FDIS	86B/3824/RVD

Full information on the voting for the approval of this standard can be found in the report on voting indicated in the above table.

This publication has been drafted in accordance with the ISO/IEC Directives, Part 2.

IEC 61753 consists of the following parts, under the general title *Fibre optic interconnecting devices and passive components – Performance standard*.

The committee has decided that the contents of this publication will remain unchanged until the stability date indicated on the IEC web site under "<http://webstore.iec.ch>" in the data related to the specific publication. At this date, the publication will be

- reconfirmed,
- withdrawn,
- replaced by a revised edition, or
- amended.

A bilingual version of this publication may be issued at a later date.

FIBRE OPTIC INTERCONNECTING DEVICES AND PASSIVE COMPONENTS – PERFORMANCE STANDARD –

Part 031-3: Non-connectorized single-mode 1×N and 2×N non-wavelength-selective branching devices for Category U – Uncontrolled environment

1 Scope

This part of IEC 61753 contains the minimum initial tests and measurement requirements and severities which a non-wavelength selective branching device (NWBD) should satisfy in order to be categorized as meeting the requirements of this standard.

The requirements cover balanced bidirectional non-connectorized single-mode $1 \times N$ and $2 \times N$ non-wavelength-selective branching devices for use in an IEC category U environment (N is the number of branching ports), especially but not exclusively used for PON application. For balanced NWBD two attenuation and uniformity performance classes are considered: class A (premium class) which meets more restrictive requirements (i.e. for extended reach PON application) and class B (standard class) for standard application (i.e. for normal reach PON application).

The requirements also cover unbalanced, bidirectional, non-connectorized, single-mode, non-wavelength-selective branching devices; however, the specifications of unbalanced branching devices are limited to 1×2 and 2×2 devices because they are the most commonly used.

2 Normative references

The following documents, in whole or in part, are normatively referenced in this document and are indispensable for its application. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 60793-2-50:2012, *Optical fibres – Part 2-50: Product specifications – Sectional specification for class B single-mode fibres*

IEC 61300-2-1, *Fibre optic interconnecting devices and passive components – Basic test and measurement procedures – Part 2-1: Tests – Vibration (sinusoidal)*

IEC 61300-2-4, *Fibre optic interconnecting devices and passive components – Basic test and measurement procedures – Part 2-4: Tests – Fibre/cable retention*

IEC 61300-2-5, *Fibre optic interconnecting devices and passive components – Basic test and measurement procedures – Part 2-5: Tests – Torsion*

IEC 61300-2-9, *Fibre optic interconnecting devices and passive components – Basic test and measurement procedures – Part 2-9: Tests – Shock*

IEC 61300-2-14, *Fibre optic interconnecting devices and passive components – Basic test and measurement procedures – Part 2-14: Tests – High optical power*

IEC 61300-2-17, *Fibre optic interconnecting devices and passive components – Basic test and measurement procedures – Part 2-17: Tests – Cold*

IEC 61300-2-18, *Fibre optic interconnecting devices and passive components – Basic test and measurement procedures – Part 2-18: Tests – Dry heat – High temperature endurance*

IEC 61300-2-22, *Fibre optic interconnecting devices and passive components – Basic test and measurement procedures – Part 2-22: Tests – Change of temperature*

IEC 61300-2-42, *Fibre optic interconnecting devices and passive components – Basic test and measurement procedures – Part 2-42: Tests – Static side load for connectors*

IEC 61300-2-44, *Fibre optic interconnecting devices and passive components – Basic test and measurement procedures – Part 2-44: Tests – Flexing of the strain relief of fibre optic devices*

IEC 61300-2-46, *Fibre optic interconnecting devices and passive components – Basic test and measurement procedures – Part 2-46: Tests – Damp heat, cyclic*

IEC 61300-3-2:2009, *Fibre optic interconnecting devices and passive components – Basic test and measurement procedures – Part 3-2: Examination and measurements – Polarization dependent loss in a single-mode fibre optic device*

IEC 61300-3-3:2009, *Fibre optic interconnecting devices and passive components – Basic test and measurement procedures – Part 3-3: Examinations and measurements – Active monitoring of changes in attenuation and return loss*

IEC 61300-3-6:2009, *Fibre optic interconnecting devices and passive components – Basic test and measurement procedures – Part 3-6: Examinations and measurements – Return loss*

IEC 61300-3-7:2009, *Fibre optic interconnecting devices and passive components – Basic test and measurement procedures – Part 3-7: Examinations and measurements – Wavelength dependence of attenuation and return loss of single mode components*

IEC 61300-3-20, *Fibre optic interconnecting devices and passive components – Basic test and measurement procedures – Part 3-20: Examinations and measurements – Directivity of fibre optic branching devices*

IEC 61300-3-28, *Fibre optic interconnecting devices and passive components – Basic test and measurement procedures – Part 3-28: Examinations and measurements – Transient loss*

3 Test

All test methods are selected within the IEC 61300 series.

The samples for tests shall be terminated onto single-mode fibres according to category B1.1, B1.3, or B.6 of IEC 60793-2-50:2012 in either coated fibres (primary and secondary) or reinforced cable format.

All tests shall be carried out to validate performance over one of the spectral bands listed below:

- 1) Spectral bands I:
 - 1 260 nm to 1 360 nm
 - 1 480 nm to 1 625 nm

2) Spectral bands II:

1 260 nm to 1 360 nm

1 480 nm to 1 660 nm

4 Test report

Fully documented test reports and supporting evidence shall be prepared and be available for inspection as evidence that the tests have been carried out and complied with.

5 Performance requirements

5.1 Dimensions

Dimensions shall comply with those given in appropriate manufacturer's drawings.

5.2 Sample size

Sample sizes for the tests are defined in Annex B.

5.3 Test details and requirements

Performance requirements and details are specified in Table 1.

All optical performances are given only for non-connectorized NWBD. During the environmental tests where monitoring of the NWBD is needed, all ports of the device shall be monitored.

In Annex A some numerical values of attenuation and uniformity requirements of test numbers 1 and 2 for the most commonly used NWBD are shown in Tables A.1, A.2 and A.3.

In Tables A.4 and A.5 the minimum attenuation requirements at room temperature are described by way of equations on the top of column, with the calculated values of the most commonly used NWBD listed below.

Table 1 – Test details and requirements (1 of 7)

No.	Tests	Requirements						Details	
1	Attenuation (A) (Insertion loss) IEC 61300-3-7:2009 (Method A)	Balanced NWBD	1 x N			2 x N		Launch patchcord length	≥ 2 m
		Configuration						Source type	Unpolarized
		Performance class	A	B		A	B	Launch conditions	The wavelength of the source shall be longer than the cut-off wavelength of the fibre
		Spectral band I	≤ 0,5 + 3,3log ₂ N (dB)	≤ 0,5 + 3,4log ₂ N (dB)		≤ 0,7 + 3,4log ₂ N (dB)	≤ 0,7 + 3,5log ₂ N (dB)		
		Spectral band II	≤ 0,5 + 3,4log ₂ N (dB)	≤ 0,5 + 3,5log ₂ N (dB)		≤ 0,7 + 3,5log ₂ N (dB)	≤ 0,7 + 3,6log ₂ N (dB)		
		Unbalanced NWBD						Uncertainty	≤ ± 0,05 dB
		Spectral band I Spectral band II	≤ 22 – 10,5log ₁₀ <i>P</i> (dB) where <i>P</i> is the nominal percentage of power associated with one port						The measurement should be performed with all combination of input/output ports See Tables A.1, A.2 and A.3 for example
2	Uniformity (U) IEC 61300-3-7:2009 (Method A)	Balanced NWBD	1 x N			2 x N		Launch patchcord length	≥ 2 m
		Configuration						Source type	Unpolarized
		Performance Class	A	B		A	B	Launch conditions	The wavelength of the source shall be longer than the cut-off wavelength of the fibre
		Spectral band I	≤ 0,1 + 0,3log ₂ N (dB)	≤ 0,2 + 0,3log ₂ N (dB)		≤ 0,4 + 0,4log ₂ N (dB)	≤ 0,5 + 0,4log ₂ N (dB)		
		Spectral band II	≤ 0,1 + 0,4log ₂ N (dB)	≤ 0,2 + 0,4log ₂ N (dB)		≤ 0,4 + 0,5log ₂ N (dB)	≤ 0,5 + 0,5log ₂ N (dB)		
								Uncertainty	≤ ± 0,05 dB
									The measurement should be performed with all combination of input/output ports See Tables A.1 and A.2 for example

Table 1 (2 of 7)

No.	Tests	Requirements	Details	
3	Directivity IEC 61300-3-20	≥ 55 dB	Launch patchcord length Source type Launch conditions Uncertainty	≥ 2 m LD The wavelength of the source shall be longer than the cut-off wavelength of the fibre. $\leq \pm 1$ dB The measurement should be made between all pairs of Input/output ports. All ports not under test shall be terminated to avoid unwanted reflections contributing to the measurement
4	Return loss (RL) IEC 61300-3-6:2008 (Method 1, OCWR)	≥ 55 dB	Launch patchcord length Source type Launch conditions Uncertainty	≥ 2 m LD The wavelength of the source shall be longer than the cut-off wavelength of the fibre. $\leq \pm 1$ dB The measurement should be performed with all combination of input/output ports. All ports not under test shall be terminated to avoid unwanted reflections contributing to the measurement
5	Polarization dependent loss (PDL) IEC 61300-3-2:2009	For balanced NWBD: $\begin{array}{ccc} & 1 \times N & 2 \times N \\ N \leq 4 & \leq 0,2 \text{ dB} & \leq 0,3 \text{ dB} \\ 4 < N \leq 16 & \leq 0,3 \text{ dB} & \leq 0,4 \text{ dB} \\ N > 16 & \leq 0,4 \text{ dB} & \leq 0,5 \text{ dB} \end{array}$ For unbalanced 1×2 and 2×2 NWBD and for any pair of input and output ports: $\leq 0,7 - 0,25 \log_{10} P$ (dB) where P is the nominal percentage of the power associated with one port	Launch patchcord length Source type Uncertainty	≥ 2 m LD $\leq \pm 0,05$ dB The measurement should be performed with all combination of input/output ports

Table 1 (3 of 7)

No.	Tests	Requirements	Details	
6	High optical power IEC 61300-2-14	$P_{\max}$ = 500 mW (+27 dBm) per port only one at the time During the test the change in A shall be within $\pm 0,5$ dB of the original value under ambient conditions, while on completion of the test the change in A shall be within $\pm 0,3$ dB of the original value During and on completion of the test the RL limit of Test No. 4 shall be met	Launch patchcord length Source type Launch conditions Uncertainty A test Uncertainty RL test Test wavelength Spectral band I Test wavelength Spectral band II Duration of the optical power exposure at the each level	≥ 2 m Unpolarized The wavelength of the source shall be longer than the cut-off wavelength of the fibre. $\leq 0,05$ Db $\leq \pm 1$ dB 1 310 nm $\pm$ 20 nm 1 550 nm $\pm$ 20 nm 1 310 nm $\pm$ 20 nm 1 625 nm $\pm$ 20 nm 30 min The measurement shall be performed between the common port (input port) and the output ports
7	Cold IEC 61300-2-17	For balanced and unbalanced NWBD before and on completion of the test the A limits of Test No. 1 shall be met. In addition, during the test the change in A of balanced NWBD shall be within $\pm 0,3$ dB for $N \leq 4$ and within $\pm 0,5$ dB for $N > 4$ of the original value under ambient conditions. For unbalanced NWBD the change in A during the test shall be within $\pm 0,3$ dB for $P \% > 2$ % and $\pm 0,5$ dB for $P \% \leq 2$ % of the original value. During and after the test the RL limit of Test No. 4 shall be met	Temperature Duration of exposure	-25 °C $\pm$ 2 °C 96 h Specimens shall be optically functioning. A and RL shall be measured before the test, during the test at a maximum interval of 1 h and after the test by the Method 3 of IEC 61300-3-3:2009
8	Dry heat – High temperature endurance IEC 61300-2-18	For balanced and unbalanced NWBD before and on completion of the test the A limits of Test No. 1 shall be met. In addition during the test the change in A of balanced NWBD shall be within $\pm 0,3$ dB for $N \leq 4$ and within $\pm 0,5$ dB for $N > 4$ of the original value under ambient conditions. For unbalanced NWBD the change in A during the test shall be within $\pm 0,3$ dB for $P \% > 2$ % and $\pm 0,5$ dB for $P \% \leq 2$ % of the original value. During and after the test the RL limit of Test No. 4 shall be met	Temperature Duration of exposure	$+70$ °C $\pm$ 2 °C 96 h Specimens shall be optically functioning. A and RL shall be measured before the test, during the test at a maximum interval of 1 h and after the test by the Method 3 of IEC 61300-3-3:2009

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No.	Tests	Requirements	Details	
9	Damp heat (cyclic) IEC 61300-2-46	For balanced and unbalanced NWBD before and on completion of the test the A limits of Test No. 1 shall be met. In addition during the test the change in A of balanced NWBD shall be within $\pm 0,3$ dB for $N \leq 4$ and within $\pm 0,5$ dB for $N > 4$ of the original value under ambient conditions. For unbalanced NWBD the change in A during the test shall be within $\pm 0,3$ dB for $P \% > 2$ % and $\pm 0,5$ dB for $P \% \leq 2$ % of the original value. During and after the test the RL limit of Test No. 4 shall be met	High temperature Low temperature Relative humidity Number of cycles Duration of each cycle	$+55\text{ }^{\circ}\text{C} \pm 2\text{ }^{\circ}\text{C}$ $+25\text{ }^{\circ}\text{C} \pm 2\text{ }^{\circ}\text{C}$ $>90\text{ } \%$ 10 24 h Specimens shall be optically functioning. A and RL shall be measured before the test, during the test at a maximum interval of 10 min and after the test by the Method 3 of IEC 61300-3-3:2009
10	Change of temperature IEC 61300-2-22	For balanced and unbalanced NWBD before and on completion of the test the A limits of Test No. 1 shall be met. In addition during the test the change in A of balanced NWBD shall be within $\pm 0,3$ dB for $N \leq 4$ and within $\pm 0,5$ dB for $N > 4$ of the original value under ambient conditions. For unbalanced NWBD the change in A during the test shall be within $\pm 0,3$ dB for $P \% > 2$ % and $\pm 0,5$ dB for $P \% \leq 2$ % of the original value. During and after the test the RL limit of Test No. 4 shall be met	High temperature Low temperature Duration at extreme temperature Temperature rate of change Number of cycles	$+70\text{ }^{\circ}\text{C} \pm 2\text{ }^{\circ}\text{C}$ $-25\text{ }^{\circ}\text{C} \pm 2\text{ }^{\circ}\text{C}$ 1 h $1\text{ }^{\circ}\text{C}/\text{min}$ 12 Specimens shall be optically functioning. A and RL shall be measured before the test, during the test at a maximum interval of 10 min and after the test by the Method 3 of IEC 61300-3-3:2009

Table 1 (5 of 7)

No.	Tests	Requirements	Details	
11	Vibration (sinusoidal) IEC 61300-2-1	For balanced and unbalanced NWBD before and on completion of the test the A limits of Test No. 1 shall be met. In addition during the test the change in A of balanced NWBD shall be within $\pm 0,3$ dB for $N \leq 4$ and within $\pm 0,5$ dB for $N > 4$ of the initial value. For unbalanced NWBD the change in A during the test shall be within $\pm 0,3$ dB for $P \% > 2$ % and within $\pm 0,5$ dB for $P \% \leq 2$ % of the initial value. During and after the test the RL limit of Test No. 4 shall be met	Frequency range Number of axes Rate of change Number of sweeps Vibration amplitude	10 Hz to 55 Hz 3 orthogonal 1 octave/min. (10 Hz to 55 Hz to 10 Hz): 15 0,75 mm NWBD shall be optically functioning. A and RL shall be measured before and after the test by means the set-ups defined in Test No. 1 and No. 4. In addition during the test the change in A shall be measured by the IEC 61300-3-28 (transient loss) at $1\ 550\ \text{nm} \pm 25\ \text{nm}$
12	Static side load IEC 61300-2-42	For balanced and unbalanced NWBD before and on completion of the test the A limits of Test No. 1 shall be met In addition on completion of the test the change in A of balanced NWBD shall be within $\pm 0,3$ dB for $N \leq 4$ and within $\pm 0,5$ dB for $N > 4$ of the initial value. For unbalanced NWBD the change in A after the test shall be within $\pm 0,3$ dB for $P \% > 2$ % and within $\pm 0,5$ dB for $P \% \leq 2$ % of the initial value. After the test the RL limit of Test No. 4 shall be met	Magnitude of the load	1 N for 1 h for reinforced cables; 0,2 N for 5 min for secondary coated fibres. For primary coated fibre the test can be omitted. The test shall be applied in two mutually perpendicular directions as permitted by the device design. NWBD shall be optically functioning. A and RL shall be measured before and after the test by means the setups defined in Tests No. 1 and No. 4

Table 1 (6 of 7)

No.	Tests	Requirements	Details	
13	Fibre/cable retention IEC 61300-2-4	For balanced and unbalanced NWBD before and on completion of the test the A limits of Test No. 1 shall be met. In addition on completion of the test the change in A of balanced NWBD shall be within $\pm 0,3$ dB for $N \leq 4$ and within $\pm 0,5$ dB for $N > 4$ of the initial value. For unbalanced NWBD the change in A after the test shall be within $\pm 0,3$ dB for $P \% > 2$ % and within $\pm 0,5$ dB for $P \% \leq 2$ % of the initial value. After the test the RL limit of Test No. 4 shall be met	Magnitude and rate of application of the tensile load. Point of application of tensile load. Duration of the test (maintaining the load)	10 N $\pm$ 1 N at a speed of 5 N/s for reinforced cable; 5 N $\pm$ 0,5 N at a speed of 0,5 N/s for secondary coated fibres; 2 N $\pm$ 0,2 N at a speed of 0,5 N/s for primary coated fibres. 0,3 m from the end of the NWBD. 120 s at 10 N; 60 s at 2 N and 5 N. NWBD shall be optically functioning. A and RL shall be measured before and after the test by means the setups defined in Tests No. 1 and No. 4
14	Flexing of strain relief of fibre optic devices IEC 61300-2-44	For balanced and unbalanced NWBD before and on completion of the test the A limits of Test No. 1 shall be met. In addition, on completion of the test the change in A of balanced NWBD shall be within $\pm 0,3$ dB for $N \leq 4$ and within $\pm 0,5$ dB for $N > 4$ of the initial value. For unbalanced NWBD, the change in A after the test shall be within $\pm 0,3$ dB for $P \% > 2$ % and within $\pm 0,5$ dB for $P \% \leq 2$ % of the initial value. After the test the RL limit of Test No. 4 shall be met	Magnitude of the load Number of cycles	5 N for reinforced cable. For primary and secondary coated fibre the test can be omitted. 30 at $\pm 90^\circ$ NWBD shall be optically functioning. A and RL shall be measured before and after the test by means the setups defined in Tests No. 1 and No. 4
15	Torsion/twist IEC 61300-2-5	For balanced and unbalanced NWBD before and on completion of the test the A limits of Test No. 1 shall be met In addition, on completion of the test the change in A of balanced NWBD shall be within $\pm 0,3$ dB for $N \leq 4$ and within $\pm 0,5$ dB for $N > 4$ of the initial value. For unbalanced NWBD the change in A after the test shall be within $\pm 0,3$ dB for $P \% > 2$ % and within $\pm 0,5$ dB for $P \% \leq 2$ % of the initial value. After the test the RL limit of Test No. 4 shall be met	Magnification and rate of application of the tensile load. Point of application of the load. Duration of the test	5 N at a speed of 0,1 N/s for reinforced cable terminated NWBD. 2 N at a speed of 0,1 N/s for coated fibre terminated NWBD. For primary coated fibre the test can be omitted. 0,2 m from the end of the NWBD. 10 cycles $\pm 180^\circ$ NWBD shall be optically functioning. A and RL shall be measured before and after the test by means the setups defined in Tests No. 1 and No. 4

Table 1 (7 of 7)

No.	Tests	Requirements	Details	
16	Shock IEC 61300-2-9	For balanced and unbalanced NWBD before and on completion of the test the A limits of Test No. 1 shall be met. In addition on completion of the test the change in A of balanced NWBD shall be within $\pm 0,3$ dB for $N \leq 4$ and within $\pm 0,5$ dB for $N > 4$ of the initial value. For unbalanced NWBD the change in A after the test shall be within $\pm 0,3$ dB for $P \% > 2 \%$ and within $\pm 0,5$ dB for $P \% \leq 2 \%$ of the initial value. After the test the RL limit of Test No. 4 shall be met	Acceleration forces Nominal duration No. of shocks	Components: 5 000 m/s² for mass $\leq 0,125$ kg. Modules: 2 000 m/s² for: 0,125 kg < mass $\leq 0,225$ Kg; 500 m/s² for: 0,225 Kg < mass ≤ 1 kg. 1 ms, half sine pulse 3 axes in 2 directions, 2 shocks per axis, 12 shocks total. NWBD shall be optically functioning. A and RL shall be measured before and after the test by means the setups defined in Tests No. 1 and No. 4

Annex A (normative)

A and U requirements of $1 \times N$ and $2 \times N$ NWBDs

A.1 Attenuation and uniformity requirements of $1 \times N$ and $2 \times N$ NWBDs calculated by the equations of Tests No.1 and 2

Tables A.1, A.2 and A.3 show the attenuation and uniformity requirements for different classes with the underlying formulas as specified in the Tests 1 and 2 of Table 1.

Tableau A.1 – Attenuation and uniformity requirements of balanced bidirectional NWBD having the most common port configurations for Class A, with the underlying formulas as specified in the Tests 1 and 2 of Table 1

N°	$1 \times N$				$2 \times N$			
	Spectral band I		Spectral band II		Spectral band I		Spectral band II	
	$A_{\max}$ (dB)	$U_{\max}$ (dB)	$A_{\max}$ (dB)	$U_{\max}$ (dB)	$A_{\max}$ (dB)	$U_{\max}$ (dB)	$A_{\max}$ (dB)	$U_{\max}$ (dB)
	$0,5+3,3 \log_2 N$	$0,1+0,3 \log_2 N$	$0,5+3,4 \log_2 N$	$0,1+0,4 \log_2 N$	$0,7+3,4 \log_2 N$	$0,4+0,4 \log_2 N$	$0,7+3,5 \log_2 N$	$0,4+0,5 \log_2 N$
2	3,8	0,4	3,9	0,5	4,1	0,8	4,2	0,9
3	5,7	0,6	5,9	0,7	6,1	1,0	6,2	1,2
4	7,1	0,7	7,3	0,9	7,5	1,2	7,7	1,4
6	9,0	0,9	9,3	1,1	9,5	1,4	9,7	1,7
8	10,4	1,0	10,7	1,3	10,9	1,6	11,2	1,9
12	12,3	1,2	12,7	1,5	12,9	1,8	13,2	2,2
16	13,7	1,3	14,1	1,7	14,3	2,0	14,7	2,4
24	15,6	1,5	16,1	1,9	16,3	2,2	16,7	2,7
32	17,0	1,6	17,5	2,1	17,7	2,4	18,2	2,9
64	20,3	1,9	20,9	2,5	21,1	2,8	21,7	3,4
128	23,6	2,2	24,3	2,9	24,5	3,2	25,2	3,9

NOTE N is the number of branching ports.

Tableau A.2 – Attenuation and uniformity requirements of balanced bidirectional NWBD having the most common port configurations for Class B, with the underlying formulas as specified in the Tests 1 and 2 of Table 1

N°	1 × N				2 × N			
	Spectral band I		Spectral band II		Spectral band I		Spectral band II	
	$A_{\max}$ (dB)	$U_{\max}$ (dB)	$A_{\max}$ (dB)	$U_{\max}$ (dB)	$A_{\max}$ (dB)	$U_{\max}$ (dB)	$A_{\max}$ (dB)	$U_{\max}$ (dB)
	$0,5+3,4 \log_2 N$	$0,2+0,3 \log_2 N$	$0,5+3,5 \log_2 N$	$0,2+0,4 \log_2 N$	$0,7+3,5 \log_2 N$	$0,5+0,4 \log_2 N$	$0,7+3,6 \log_2 N$	$0,5+0,5 \log_2 N$
2	3,9	0,5	4,0	0,6	4,2	0,9	4,3	1,0
3	5,9	0,7	6,0	0,8	6,2	1,1	6,4	1,3
4	7,3	0,8	7,5	1,0	7,7	1,3	7,9	1,5
6	9,3	1,0	9,5	1,2	9,7	1,5	10,0	1,8
8	10,7	1,1	11,0	1,4	11,2	1,7	11,5	2,0
12	12,7	1,3	13,0	1,6	13,2	1,9	13,6	2,3
16	14,1	1,4	14,5	1,8	14,7	2,1	15,1	2,5
24	16,1	1,6	16,5	2,0	16,7	2,3	17,2	2,8
32	17,5	1,7	18,0	2,2	18,2	2,5	18,7	3,0
64	20,9	2,0	21,5	2,6	21,7	2,9	22,3	3,5
128	24,3	2,3	25,0	3,0	25,2	3,3	25,9	4,0

NOTE N is the number of branching ports.

Tableau A.3 – Attenuation requirements of 1 × 2 and 2 × 2 unbalanced NWBD having the most common port configurations, with the underlying formula as specified in Test 1 of Table 1

Coupling ratio $P_1 \% / P_2 \%$	$A_{1\max} / A_{2\max}$ dB
	$(22 - 10,5 \log_{10} P_1) / (22 - 10,5 \log_{10} P_2)$
40/60	5,2 / 3,3
30/70	6,5 / 2,6
20/80	8,3 / 2,0
10/90	11,5 / 1,5
5/95	14,7 / 1,2
2/98	18,8 / 1,1
1/99	22,0 / 1,0

A.2 Minimum requirements at room temperature of attenuation values for balanced bidirectional 1 × N and 2 × N NWBD

Tables A.4 and A.5 show the minimum requirements at room temperature of attenuation values for different classes.

Tableau A.4 – Minimum requirements at room temperature of attenuation values for Class A balanced bidirectional NWBD

N°	1 × N		2 × N	
	Spectral band I	Spectral band II	Spectral band I	Spectral band II
	$A_{\max}$ (dB) $0,5+3,2 \log_2 N$	$A_{\max}$ (dB) $0,5+3,3 \log_2 N$	$A_{\max}$ (dB) $0,7+3,3 \log_2 N$	$A_{\max}$ (dB) $0,7+3,4 \log_2 N$
2	3,7	3,8	4,0	4,1
3	5,6	5,7	5,9	6,1
4	6,9	7,1	7,3	7,5
6	8,8	9,0	9,2	9,5
8	10,1	10,4	10,6	10,9
12	12,0	12,3	12,5	12,9
16	13,3	13,7	13,9	14,3
24	15,2	15,6	15,8	16,3
32	16,5	17,0	17,2	17,7
64	19,7	20,3	20,5	21,1
128	22,9	23,6	23,8	24,5
NOTE N is the number of branching ports.				

Tableau A.5 – Minimum requirements at room temperature of attenuation values for Class B balanced bidirectional NWBD

N	1 × N		2 × N	
	Spectral band I	Spectral band II	Spectral band I	Spectral band II
	$A_{\max}$ (dB) $0,5+3,3 \log_2 N$	$A_{\max}$ (dB) $0,5+3,4 \log_2 N$	$A_{\max}$ (dB) $0,7+3,4 \log_2 N$	$A_{\max}$ (dB) $0,7+3,5 \log_2 N$
2	3,8	3,9	4,1	4,2
3	5,7	5,9	6,1	6,2
4	7,1	7,3	7,5	7,7
6	9,0	9,3	9,5	9,7
8	10,4	10,7	10,9	11,2
12	12,3	12,7	12,9	13,2
16	13,7	14,1	14,3	14,7
24	15,6	16,1	16,3	16,7
32	17,0	17,5	17,7	18,2
64	20,3	20,9	21,1	21,7
128	23,6	24,3	24,5	25,2
NOTE N is the number of branching ports.				

Annex B (normative)

Sample size

As shown in Table B.1, all samples shall be subjected to Tests 1 to 5. All other tests shall be carried out in any of the following order. Consecutive testing on the same optical sample is allowed, but in case of failure during the consecutive testing, a new sample shall be prepared and the failed test shall be redone.

Table B.1 – Sample size for each test

Test number	Test	Sample size
1	Attenuation (insertion loss)	12
2	Uniformity	12
3	Directivity	12
4	Return loss	12
5	Polarization dependent loss	12
6	High optical power	8
7	Cold	4
8	Dry heat – High temperature endurance	4
9	Damp heat (cyclic)	4
10	Change of temperature	4
11	Vibration	4
12	Static side load	4
13	Fibre/cable retention	4
14	Flexing of strain relief of fibre optic devices	4
15	Torsion/twist	4
16	Shock	4

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